MEMORY DIVERGENCE IS NOT BEHAVIORAL DIVERGENCE: STAGE-RESOLVED TRANSPORT IN SELF-IMPROVING AGENT MEMORY

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ABSTRACT

Self-improving agents increasingly convert terminal feedback into persistent memory, but common evaluations conflate four distinct events: a feedback intervention changes stored state, that state is retrieved, the policy uses the retrieved difference, and behavior changes. We study this transport problem with frozen success-versus-failure writer interventions on otherwise identical trajectories. Reward-conditioned writing produces different persistent memories for all 20 Shopping pairs and all 4 Reddit pairs, and a stronger same-mode wording control leaves a significant reward-mode excess. When branch-specific memory is mechanically supplied, it has measurable downstream leverage (terminal $|\Delta| = 0.15625$, $p = 0.00074$). Under native reuse, however, Shopping still retrieves branch-specific memory in 125/172 opportunities while first-action uptake is weak (TV $= 0.06944$, $p = 0.5801$, 0/36 modal changes) and terminal transport is sparse ($|\Delta| = 0.02083$, $p = 0.4289$, 34/36 zero). Reddit repeats the write-stage effect but has sparse, sign-heterogeneous native terminal differences. Because write distance, retrieval rate, action TV, and endpoint differences are not commensurate, we introduce an ordinal stage-evidence ladder that preserves each stage's own evidence semantics and localizes the first unsupported native stage after supported/observed prerequisites. It places the strongest supported attenuation after exposure and before stable action uptake. This is an operational localization, not a causal mediation estimate. The result establishes a measurement principle for persistent self-improvement: state divergence is not behavioral authority, and memory availability is not memory use.

1 INTRODUCTION

A persistent-memory agent can disagree with itself long before task accuracy reveals anything. Consider two copies of the same successful interaction trajectory. If one copy is labeled successful and the other failed before reflection, an outcome-conditioned memory writer may store different reusable advice even though the underlying experience is unchanged. That is already a meaningful state intervention. But it leaves a harder question unanswered: *where, if anywhere, does the resulting state difference become a behavioral difference?*

This distinction matters because persistent self-improvement contains several causal surfaces that are often collapsed into one "memory helped" statement. A terminal signal may first alter what is **written**; the resulting item may or may not be **exposed** by retrieval on a later task; retrieved branch-specific content may or may not be **used** by the policy; and only then can it change the terminal **outcome**. A positive memory-similarity result identifies the first surface, while a retrieval hit only identifies availability. Neither is evidence by itself that the agent's decision rule changed. Conversely, an endpoint null does not reveal whether the update failed to write, failed to retrieve, or was filtered after retrieval.

We study this problem through a stage-resolved intervention rather than a new memory algorithm. For a frozen trajectory τ, we intervene only on the success/failure writer branch and construct paired persistent memories. We then instrument four native stages,

$$\text{write} \rightarrow \text{retrieval exposure} \rightarrow \text{first-action uptake} \rightarrow \text{terminal outcome},$$

and add a separate forced fixed-evidence control that bypasses native retrieval. The forced arm is deliberately *not* another stage in the chain: it asks whether branch-specific memory content has downstream leverage when supplied, whereas the native path asks whether the deployed memory system actually transports that difference into behavior.

This design distinguishes several explanations that an endpoint-only experiment cannot separate. If the writer does not create a persistent intervention, the story stops at write time. If the downstream policy is globally insensitive to memory content, even forced exposure should have little leverage. If the only problem is retrieval absence, native exposure should collapse. If retrieval itself is a valid proxy for policy use, observed exposure should co-occur with a supported action-level branch contrast. Finally, if reward-conditioned memory has a universal directional effect, terminal differences should not be dominated by zeros and sign reversals across tasks. Because these stages have different units and experimental surfaces, we do not convert them into a single "transport efficiency." Instead, we introduce an ordinal stage-evidence ladder: preserve each stage's own inferential/descriptive semantics, then localize the first native stage whose branch contrast is no longer supported after the preceding prerequisites have been supported or directly observed.

The frozen evidence yields a sharp pattern. Reward-conditioned writing changes durable memory in all 20 Shopping pairs and all 4 Reddit pairs. A same-mode wording control shows that the reward/reflection-mode contrast exceeds a stronger within-mode wording perturbation. Forced fixed-evidence exposure produces terminal $|\Delta| = 0.15625$ ($p = 0.00074$), demonstrating downstream leverage under supplied memory. Yet native Shopping retrieval remains common—125 hits in 172 held-out opportunities—while first-action TV is only 0.06944 ($p = 0.5801$) with 0/36 modal action changes, and native terminal $|\Delta|$ is 0.02083 ($p = 0.4289$) with 34/36 zero cells. Reddit again shows reliable write divergence but sparse terminal transport: 6/8 cells are zero and the two nonzero effects have opposite signs.

Taken together, these measurements weaken three simple explanations at once: the native null is not caused by failure to create persistent state; it is not well described as global downstream insensitivity to supplied memory content; and it is not explained by retrieval absence alone. The strongest supported *operational* localization is therefore after exposure and before stable action uptake. We intentionally stop short of a causal mediation claim because the frozen pool lacks per-atom decoding-seed binding and same-condition noise-floor replication, and the forced and native surfaces are not identical interventions.

Our contributions are:

1. **A same-trajectory stage-resolved intervention for reward-conditioned memory transport.** Rather than claiming that retrieval-versus-reuse separation is itself new, we hold the source trajectory fixed, intervene on the success/failure writer branch, and follow the resulting persistent-state contrast through native exposure, first-action uptake, and terminal outcome.

2. **A capacity-versus-transport control.** Forced fixed-evidence exposure establishes that branch-specific memory can have downstream leverage, while native reuse measures whether the deployed retrieval-and-policy path realizes that leverage.

3. **An ordinal localization rule rather than a post-hoc coefficient.** We preserve stage-specific evidence states (supported, directly observed, or not supported) and define the operational bottleneck as the first unsupported native stage after supported/observed prerequisites. On the frozen evidence this yields the exposure-to-uptake boundary without inventing a cross-stage ratio or causal mediation coefficient.

4. **A cross-domain boundary rather than a universal effect claim.** Reddit reproduces write-stage divergence but shows sparse and sign-heterogeneous native terminal effects, ruling out a simple monotone "more memory difference means more behavioral difference" story on the frozen evidence.

The resulting claim is deliberately narrower than a new memory method: **persistent-state divergence is not behavioral authority, and retrieval availability is not policy use**. This boundary tells us what a self-improving agent evaluation must measure before crediting a persistent update with downstream control.

2 STAGE-RESOLVED MEMORY TRANSPORT

2.1 CONTROLLED WRITE INTERVENTION

Let τ denote a frozen interaction trajectory before persistent memory construction. We create two writer conditions on the same τ,

$$r \in \{S, F\}, \qquad m^{(r)} = g(\tau, r),$$

where g is the released memory writer and S/F selects its success- or failure-reflection mode. The operational write contrast is

$$W(\tau) = d_{\text{mem}}\Big(m^{(S)}, m^{(F)}\Big) .$$

$W > 0$ establishes that the frozen writer produced different persistent states. It does *not* establish that every textual atom is a pure causal effect of the reward bit, because decoding randomness is not separately noise-calibrated at the atom level. We therefore use the same-mode wording control in Section 4 to test the stronger alternative that ordinary prompt wording alone explains the observed write divergence.

2.2 NATIVE TRANSPORT IS A SEQUENCE OF DISTINCT OBSERVABLES

For a held-out future query q, define native retrieval exposure

$$E_r(q) = \mathbf{1}[\text{Retrieve}(q; \mathcal{M}^{(r)}) \neq \varnothing].$$

Exposure is an availability variable: it says that branch-conditioned persistent state reached the policy context. It does not say that the policy used the branch-specific part of that state.

We therefore measure the first downstream decision boundary separately. Let $p_r^{(1)}(\cdot \mid q)$ be the first-action distribution under branch r. The action-level contrast is

$$U(q) = \text{TV}\Big(p_S^{(1)}(\cdot \mid q), p_F^{(1)}(\cdot \mid q)\Big) .$$

Finally, for terminal outcome $y_r(q)$,

$$O(q) = |y_S(q) - y_F(q)| .$$

The native scientific object is therefore the ordered observation tuple

$$\mathcal{T}_{\text{native}}(q) = \big(W, E, U, O\big),$$

not any one component in isolation.

Crucially, these components are **not commensurate**. W is a memory-state distance, E is an exposure indicator/rate, U is a distributional action distance, and O is an endpoint contrast. We therefore do not divide them into a scalar “transport efficiency” or treat their numerical differences as a mediation decomposition. The order is mechanistic; the scales are not.

2.3 FORCED LEVERAGE IS A CAPACITY CONTROL, NOT A NATIVE STAGE

The forced fixed-evidence experiment mechanically supplies paired memory content to the downstream task packet. Let

$$L_{\text{forced}}(q) = \left|y^{\text{forced},S}(q) - y^{\text{forced},F}(q)\right| .$$

Because native retrieval is bypassed, L_{forced} is not inserted between W and E. It answers a different question: *can* branch-specific memory content affect the downstream system when supplied? Native (E, U, O) asks whether the deployed memory path *does* transport that contrast. This capacity-versus-transport separation is the key control that prevents a weak native endpoint from being summarized as “memory can never matter.”

2.4 AN ORDINAL STAGE-EVIDENCE LADDER

The central measurement problem is that W, E, U, and O cannot be compared on one numerical scale. We therefore localize attenuation with an *ordinal evidence ladder* rather than a post-hoc transport coefficient. Let the ordered native stages be

$$s_1 = \text{write}, \quad s_2 = \text{retrieval exposure}, \quad s_3 = \text{first-action uptake}, \quad s_4 = \text{terminal outcome}.$$

For each stage we preserve its native evidence semantics. A stage may be SUPPORTED by its frozen inferential test, OBSERVED directly when the claim is descriptive rather than thresholded, or NOT-SUPPORTED when its frozen primary test does not establish a stable branch contrast. We then define

$$b^* = \min\{j : s_j \text{ is not supported/observed while } s_1, \ldots, s_{j-1} \text{ are supported/observed}\}.$$

This is an ordering rule, not a ratio: no cross-stage units are divided, normalized, or interpreted as mediation mass.

The forced fixed-evidence arm is evaluated separately as a *side capacity condition*. It can weaken the global-insensitivity explanation, but because it bypasses native retrieval it cannot make an otherwise unsupported native stage pass. This prevents the positive forced endpoint from being inserted into the native chain merely because it is statistically strong.

The same ladder can be read as an alternative-explanation audit:

$$\begin{aligned}
\textbf{A1: no state intervention} &\Rightarrow W \approx 0, \\
\textbf{A2: global downstream insensitivity} &\Rightarrow L_{\text{forced}} \approx 0, \\
\textbf{A3: retrieval absence only} &\Rightarrow E \approx 0, \\
\textbf{A4: exposure is uptake} &\Rightarrow \text{observed } E \text{ should be accompanied by supported } U, \\
\textbf{A5: universal directional transport} &\Rightarrow O \text{ should be broadly nonzero with stable sign.}
\end{aligned}$$

In our frozen evidence, write is SUPPORTED, native exposure is OBSERVED, first-action uptake is NOT-SUPPORTED, and terminal transport is sparse and sign-heterogeneous. Hence $b^* = s_3$: the strongest supported operational attenuation boundary is **after exposure and before stable first-action uptake**. The positive forced side control further weakens a global-insensitivity account without changing b^*.

This is still not an identified mediator coefficient. A causal mediation statement would additionally require compatible intervention surfaces and a noise/seed design that isolates each branch-specific atom. We do not have that evidence and do not make that claim.

2.5 INTERPRETATION DISCIPLINE

Three non-implications are load-bearing throughout the paper:

$$\text{persistent-state divergence} \not\Rightarrow \text{behavioral divergence},$$

$$\text{retrieval exposure} \not\Rightarrow \text{policy uptake},$$

$$\text{forced leverage} \not\equiv \text{native end-to-end transport}.$$

These distinctions also explain why our stopped evidence-gated residual extension is not a negative method result: that extension failed qualification before a behavioral method-effect experiment was authorized. The present paper therefore remains an identification/measurement study of the transport chain.

3 STAGE 1: REWARD-CONDITIONED WRITING ROBUSTLY CHANGES PERSISTENT STATE

We first test the write stage with paired interventions over released WebArena Shopping trajectories. Within each pair, the task text and full action history are fixed before the writer is called. One branch uses the released success-reflection objective and the other uses the released failure-reflection objective; the resolved writer model is matched within pair and temperature is zero. Raw responses and the pre-writer trajectory projection are content-addressed before analysis.

The initial F0 tranche requested six source trajectories. Four returned complete memory outputs in both conditions; the two incomplete requests terminated in the failure-label writer before assistant text existed and were excluded from the paired content estimand. Among the four complete pairs, every pair changed normalized memory content and the extracted memory-item title set, with mean token-set Jaccard distance 0.735. Receipt-level forensics preserve the two incomplete arms as provider-response failures rather than treating them as negative memory examples.

The later frozen breadth tranche expands the write-stage evidence to 20 complete Shopping success/failure pairs. All 20/20 pairs produce different branch memories, with pooled token-set Jaccard distance 0.673. When combined with the four Reddit pairs used in the cross-domain replication, the D0 lineage contains 24/24 paired source experiences with the same pre-writer trajectory projection, the same resolved writer model within pair, temperature zero, and different branch-memory content. This establishes a robust operational branch contrast at the persistent-state level.

The content changes are not only token-level. A deterministic post-ready strategy-slot audit maps frozen memory titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification slots. In the original four complete F0 pairs, the strategy-slot set changes in all four. We treat this as a structural diagnostic rather than a semantic-validity claim: the current data do not certify each textual delta atom as a pure reward-caused proposition because explicit decoding seeds or same-condition noise-floor replications are unavailable.

The write-stage result therefore supports a narrow statement: under the released writer policy, flipping the reward-conditioned reflection branch on the same source experience reliably produces a different durable state. It does not by itself establish that the difference will be retrieved, followed, or reflected in terminal behavior. Section 4 first tests the strongest simple prompt-sensitivity reduction; Section 5 then measures downstream transport stage by stage.

4 Ruling Out Generic Prompt Sensitivity

The reward intervention changes the reflection instruction together with the reward semantics. A generic prompt-sensitivity account therefore predicts that a sufficiently strong wording perturbation inside one reflection mode should produce comparable memory divergence. We test that reduction on eight released trajectories disjoint from F0 and balanced across original outcomes.

Before provider calls, we freeze semantic-preserving rewordings for both reflection modes. Under token-set Jaccard distance, the released success-versus-failure prompts differ by 0.114. The within-success rewording differs by 0.171 and the within-failure rewording by 0.152. The control thus perturbs surface wording more strongly while preserving the reflection objective.

Each fresh trajectory is written with the released success mode and the released failure mode. The same writer also processes one frozen semantic-preserving rewording of each mode at temperature zero. Let B_i be the memory distance between the two released reward modes and let W_i be the mean within-mode rewording distance. The frozen paired statistic is

$$\Delta_{\mathrm{prompt}} = \frac{1}{8}\sum_{i=1}^{8}(B_i - W_i). \tag{1}$$

All 32 control outputs complete. Mean reward-mode distance is 0.700, while the stronger same-mode wording control has mean 0.595. The paired excess is 0.105, and an exact one-sided sign-flip test gives $p = 0.0078$. This passes the frozen 0.10 effect margin together with the $p < 0.05$ gate. The reward-conditioned write effect therefore survives a same-information prompt-wording reduction that is deliberately stronger in lexical perturbation.

5 Transport Bottleneck Localization

The write intervention becomes scientifically useful only once alternative explanations for downstream attenuation are separated. We therefore use the frozen evidence in three roles: forced

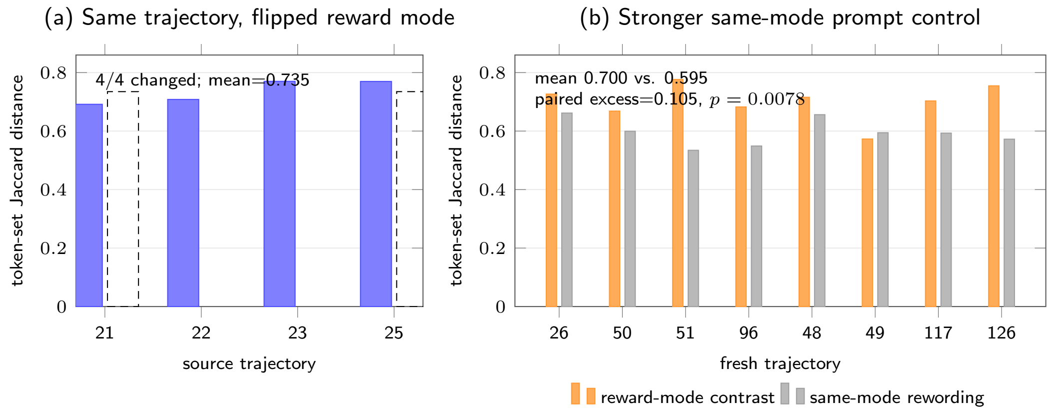

Figure 1: Write-channel evidence and its strongest simple control. (a) Among the four F0 pairs complete in both writer conditions, flipping only the reward-conditioned writer mode changes every memory. (b) On eight fresh control trajectories, reward-mode memory divergence exceeds stronger semantic-preserving same-mode prompt rewording. The control mean is 0.595 versus 0.700 between reward modes; paired excess is 0.105 with exact $p = 0.0078$.

fixed-evidence exposure as a *capacity control*, native retrieval as an *availability measurement*, and first-action/terminal contrasts as *native transport measurements*. This section shows that their joint pattern is more informative than any endpoint alone.

5.1 Forced injection establishes downstream capacity, not native transport

We first bypass native retrieval and directly inject each paired memory into a fixed future-task packet. This asks whether the downstream system can respond to the paired persistent states when the experimenter supplies them.

The original fixed-state probe contains 12 aligned success-memory versus failure-memory rollouts; 7/12 choose different structured next actions. We then freeze all four complete initial F0 source-memory pairs and four future tasks before terminal-policy execution. Every source pair is crossed with every future task, giving 16 cells. The confirmatory run uses eight fresh rollouts per memory condition in every cell, for 256 completed calls. Mean absolute terminal success-rate difference is 0.15625, above the frozen 0.15 practical floor, and the 100,000-draw within-cell permutation test gives $p = 0.00074$.

This result weakens the explanation that the downstream system is simply incapable of responding to memory content. It does *not* estimate native transport because exposure is mechanically supplied. The cell effects are also heterogeneous: eight cells are zero, six have positive failure-minus-success effects, and two have negative effects; the two largest cells account for 78.2% of squared-effect mass. Even under forced exposure, a universal harmful direction is unsupported.

5.2 Native exposure remains common while action uptake attenuates

We next restore the native memory-bank pathway. After expanding the frozen Shopping memory bank, branch-specific source memories are retrieved on 125 of 172 held-out retrieval opportunities, a rate of 0.727. The native null therefore cannot be summarized as “the memory was never retrieved.”

Despite that exposure, the first measured decision boundary is weak. Across the frozen 36 Shopping branch-comparison states, mean first-action total-variation distance is 0.06944 with $p = 0.5801$, and 0/36 states change their modal first action. A post-hoc working-memory diagnostic yields branch shift 0.00335 with $p = 0.2052$. Native terminal transport is also small: mean absolute success difference is 0.02083 with permutation $p = 0.4289$, and 34/36 cells are exactly zero.

The conjunction is the mechanism result. Applying the ordinal rule from Section 2.4, write is Supported, native exposure is Observed, and first-action uptake is the first native stage whose

Table 1: Stage-evidence ladder. Evidence states preserve the semantics of each measurement instead of forcing heterogeneous quantities onto one scale. The forced arm is a side capacity control and therefore does not enter the native-stage ordering.

Stage	Evidence state	Frozen observation / interpretation
Persistent write	Supported	20/20 Shopping pairs diverge; stronger same-mode control leaves excess 0.105, exact $p = 0.0078$.
Forced capacity (side control)	Supported	$\lvert\Delta\rvert = 0.15625$, $p = 0.00074$; supplied memory has downstream leverage but native retrieval is bypassed.
Native exposure	Directly observed	125/172 held-out opportunities retrieve branch-specific source memory; this is availability, not policy use.
First-action uptake	Not supported	TV $= 0.06944$, $p = 0.5801$, 0/36 modal changes; this is the first unsupported native stage after observed prerequisites.
Native outcome	Sparse / heterogeneous	Shopping 34/36 zero; Reddit 6/8 zero and the two nonzero signs are opposite.

Table 2: Alternative-explanation audit. The contribution comes from the joint pattern, not from selecting only the positive forced endpoint. "Weakened" means the frozen evidence makes the explanation incomplete but does not mathematically eliminate every variant of it.

Explanation	Frozen evidence	Disposition
No persistent state intervention	20/20 Shopping and 4/4 Reddit writes diverge; same-mode excess $= 0.105$, $p = 0.0078$	Inconsistent with the observed write-stage intervention.
Downstream policy is globally insensitive to memory	Forced terminal $\lvert\Delta\rvert = 0.15625$, $p = 0.00074$	Weakened: supplied branch-specific memory has measurable leverage.
Native null is only retrieval absence	125/172 native Shopping retrieval hits	Weakened: substantial exposure coexists with weak first-action contrast.
Retrieval hit is a surrogate for policy use	first-action TV $= 0.06944$, $p = 0.5801$; 0/36 modal changes	Rejected as an evaluation equivalence: availability is not uptake.
Universal directional terminal transport	Shopping 34/36 zero; Reddit 6/8 zero, two nonzero signs opposite	Not supported by the frozen native cells.
Causal mediation has been identified	no per-atom seed binding/noise-floor replication; forced/native surfaces differ	Unresolved and explicitly outside the claim set.

frozen primary test does not support a stable branch contrast. Thus $b^* =$ first-action uptake, which places the strongest supported operational attenuation boundary after exposure and before stable action uptake. Terminal sparsity is downstream corroboration rather than the quantity used to choose the boundary. We do not call this a mediation effect because the stage measurements are not commensurate coefficients and the frozen design lacks the seed/noise-floor controls required for atom-level causal mediation.

5.3 Cross-domain boundary: write replication without monotone transport

We repeat the write-versus-native-transport measurement on Reddit. Reward-conditioned writing again diverges in 4/4 paired source memories. Native terminal transport is larger than Shopping in magnitude but remains unresolved and highly sparse: mean absolute success difference is 0.125 with $p = 0.2253$, 6/8 cells are zero, and the two nonzero cells have opposite signs. The second domain therefore supports the same structural separation—robust write divergence need not become stable

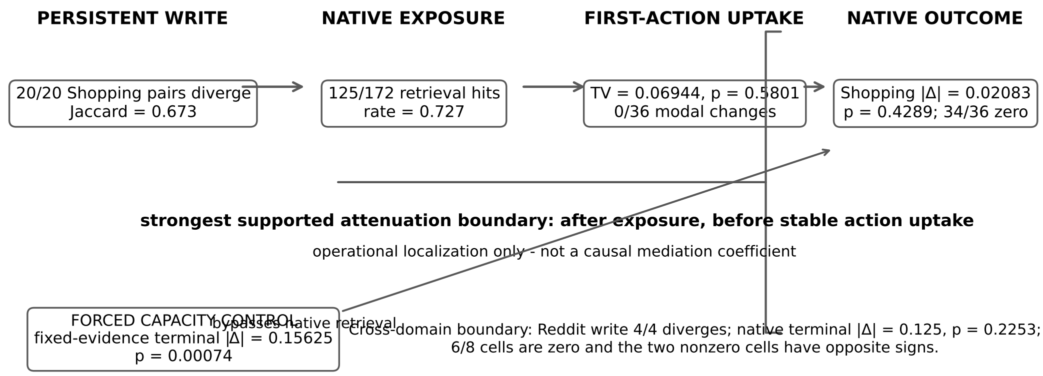

Figure 2: **Native transport and forced capacity are different experimental surfaces.** The horizontal path is the deployed reuse chain: persistent write $\rightarrow$ native exposure $\rightarrow$ first-action uptake $\rightarrow$ native outcome. The forced fixed-evidence arm bypasses retrieval and is shown as a side capacity control. The strongest supported operational attenuation boundary is after exposure and before stable action uptake; this is not a causal mediation coefficient.

native terminal transport—while rejecting a simple universal attenuation constant or signed harm coefficient.

An outcome-blind structured-memory control gives a complementary warning about endpoint interpretation. It produces a small terminal contrast of 0.04514 ($p = 0.0048$), below the frozen 0.15 practical floor, with 32/36 cells zero. Statistical detectability alone is therefore not the scientific target; a sparse small effect can pass a null test without matching the magnitude or mechanism of forced branch leverage.

5.4 Why the one-step reward-to-behavior story fails

A one-step interpretation would take the 256 forced rollouts as evidence that "reward error propagates to future behavior." The native measurements show why this is too coarse. Forced injection estimates leverage conditional on making a particular memory state available; native transport additionally depends on retrieval competition, context composition, and policy uptake. The difference between 0.15625 forced leverage and 0.02083 native Shopping terminal transport is therefore not a scalar attenuation factor to divide or average. It is evidence that two experimental surfaces answer different questions.

Conversely, looking only at the native endpoint would also lose the mechanism: a 0.02083 Shopping terminal difference does not tell us that the writer failed, that memory was unavailable, or that the downstream system was globally insensitive. The stage-resolved controls exclude or weaken each of those simpler accounts before the first-action boundary is reached.

The evaluation consequence is concrete. For a persistent update, report at least one state-change metric, one exposure measurement, and one downstream uptake/outcome metric; use a capacity control only if its bypassed stages are explicit. Then localize the first point where evidence for transport weakens, rather than crediting the update with behavioral authority because its representation changed or because a retrieved item appeared in context.

6 RELATED WORK

Reward and judge reliability. JudgeBench and RewardBench establish that LLM judges and reward models can be unreliable before their outputs are used for learning (Tan et al., 2025; Lambert et al., 2024); Plan-RewardBench shows additional brittleness on tool-integrated trajectories (Wang et al., 2026). Those works evaluate the scoring surface. We ask what happens after such a score controls a persistent state update.

Outcome-conditioned and learned memory construction. ReasoningBank explicitly distills different memories from successful and failed trajectories (Ouyang et al., 2026), Reflexion stores evaluative verbal feedback for later trials (Shinn et al., 2023), and Agent Workflow Memory extracts reusable procedures from interaction histories (Wang et al., 2024). Mem-α learns how to construct external memory from downstream reward (Wang et al., 2025), while AttriMem augments outcome reward with attribution-derived process feedback (Li et al., 2026a). We therefore do not claim feedback-driven memory construction as new. Our intervention freezes the writer policy and uses the success/failure branch only as a controlled state-update surface.

Memory valuation, retrieval, and post-retrieval reuse. Memory Reward Inflation studies how imperfect proxy reward can be amplified as memories are scored and reused (Asadolahi et al., 2026). RoMeRL studies misleading trajectory-level utility updates and the memory-reward trap (Yang et al., 2026). Most directly, Beyond Retrieval / QCR argues that retrieving a relevant trajectory and using it effectively after retrieval are distinct problems; its evaluation holds retrieval candidate, target state, model, decoding, and tool budget fixed while varying the support delivered to the acting agent (Li et al., 2026b). We therefore do *not* claim the retrieval-versus-reuse distinction as novel. Our residual object starts from a different intervention: the source trajectory itself is frozen while the reward-conditioned writer branch changes persistent state, after which we jointly measure write, native exposure, first-action uptake, and endpoint transport, with a forced-capacity side control and an ordinal first-unsupported-stage rule.

Variance and longitudinal fragility. Ye et al. show run-to-run variance, curriculum sensitivity, and memory underspecification in self-improving agents (Ye et al., 2026). Their results motivate measuring downstream instability, but variance alone does not localize where an update enters the system. Our forced/native contrast separates latent behavioral leverage from end-to-end native transport and shows that the largest attenuation can occur after a persistent state has already been created.

Evaluation boundary. Memory-lifecycle decompositions already separate phases such as write, store, retrieve, execute, propagate, and rollback (Lin et al., 2026); QCR likewise separates retrieval from post-retrieval reuse (Li et al., 2026b). We therefore do not claim that stage names or retrieval-versus-use are new. Our residual contribution is a controlled evaluation object: the same source trajectory is held fixed while reward-conditioned writing creates paired persistent states, forced fixed-evidence exposure probes downstream capacity on a bypassed surface, and native write/exposure/action/outcome evidence is interpreted with an ordinal first-unsupported-stage rule. Robust writing weakens a no-state-intervention account; forced leverage weakens global downstream insensitivity; substantial native exposure weakens retrieval absence; weak first-action change rejects retrieval-as-uptake as an evaluation equivalence; and sparse sign-heterogeneous endpoints rule out a simple universal directional transport story on the frozen evidence. This is why the positive and null results must be reported together rather than compressed into one endpoint or one cross-stage coefficient.

7 LIMITATIONS

Our intervention identifies an operational success-versus-failure writer contrast, not atom-level causal purity for every generated sentence. The original incomplete failure-label provider calls remain excluded from paired text estimands because no assistant text exists for those arms. More broadly, the current frozen pool has no explicit decoding-seed binding or same-condition same-trajectory noise-floor replication for each memory atom. We therefore avoid claiming that every textual delta is individually caused by the reward bit.

The stage measurements also use different experimental surfaces and scales. Forced fixed-evidence injection deliberately bypasses native retrieval and should be interpreted only as downstream capacity/leverage. Native Shopping and Reddit estimates include the system's retrieval and policy-uptake pathway. Memory distance, retrieval rate, first-action TV, and terminal $|\Delta|$ cannot be divided into a meaningful transport-efficiency or mediation coefficient. Our exposure-to-uptake bottleneck is therefore an operational localization supported by an implication audit, not an identified causal mediator.

The alternative-explanation audit is asymmetric. Robust writing makes a no-state-intervention account inconsistent with the frozen evidence, but forced leverage and native exposure only *weaken* broader classes of global-insensitivity and retrieval-failure explanations; they do not eliminate every possible version of those hypotheses. Similarly, the weak first-action gate does not prove that no later latent policy state ever changes. It identifies the first measured decision boundary at which stable branch transport is not supported.

The native transport effects are heterogeneous and sparse. In Shopping, 34/36 terminal cells are zero; in Reddit, 6/8 are zero and the two nonzero cells have opposite signs. The data therefore do not support a universal directional effect, a universal attenuation factor, or a model-independent effect size. The current writer/task evidence is also limited in model and domain breadth. Source-faithful live WebArena browser navigation remains unavailable in the released environment, so the native experiments use the frozen supported pathway rather than a newly reconstructed live endpoint.

Finally, our attempted evidence-gated residual method extension did not reach behavioral evaluation. It stopped at method realization because the frozen support contained no independently qualified outcome-independent semantic-validity adjudicator. This is not evidence that the method effect is negative, nor does it invalidate the stage-resolved measurements. We report this boundary to avoid turning a qualification failure into an unrun method claim.

8 Conclusion

Reward-conditioned writing is a strong persistent-state intervention, but state persistence and behavioral transport are different scientific objects. On the frozen same-trajectory controls, the writer reliably creates branch-specific memory. A separate forced fixed-evidence experiment shows that supplied branch-specific memory has downstream leverage. Yet under native reuse, branch-specific memories are still retrieved frequently while the first measured action contrast is weak and terminal differences are sparse and heterogeneous.

The joint pattern matters more than any one number. Under the ordinal evidence ladder, write is supported, native exposure is directly observed, and first-action uptake is the first native stage whose frozen primary test does not support a stable branch contrast. The forced capacity arm separately weakens global downstream insensitivity, while terminal sparsity and sign heterogeneity constrain downstream generalization. Subject to the stated causal-purity limitations, the strongest supported localization is therefore **after exposure and before stable first-action uptake**.

This changes the evaluation target for persistent self-improvement. The relevant object is not simply whether memory changed or whether a retrieved item appeared in context, but whether a controlled state difference survives

$$\text{write} \rightarrow \text{exposure} \rightarrow \text{uptake} \rightarrow \text{outcome}.$$

A capacity control may show that memory *can* matter, while the native chain determines whether the deployed system actually lets it matter. The stage-evidence ladder turns this distinction into a reusable evaluation protocol without pretending heterogeneous observables share a unit. For the released ReasoningBank-based configuration, that protocol localizes the first unsupported native stage at action uptake: **persistent-state divergence is not behavioral authority, and availability is not use**. Future memory systems should be evaluated at the stage where state differences become decisions, not credited with behavioral improvement at the stage where they are merely written or retrieved.

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A F0 REPRODUCTION DETAILS

A.1 RELEASED TRAJECTORY SAMPLE

The F0 sample uses six task IDs from the released AWM shuffle-seed-42 Shopping parquet: 21 through 25, plus 47. The deterministic sampler selects the lowest task IDs with parseable trajectory records within each original-outcome stratum. Original failures are 21 through 22 plus 24. Original successes are 23 plus 25 and 47.

A.2 MEMORY CONSTRUCTION INTERVENTION

Each released trajectory is summarized into an action trace. The same trace is inserted into two prompts. One prompt contains the released ReasoningBank success-reflection system instruction. The paired prompt contains the released failure-reflection system instruction. The task prompt stays fixed. The writer model stays fixed. Temperature stays at zero. The raw provider output is archived before metric computation.

Table 3: Complete paired interventions.

Task	Original outcome	Content changed	Token distance
21	failure	yes	0.691489
22	failure	yes	0.708108
23	success	yes	0.769953
25	success	yes	0.769608

A.3 Pair-level F0 results

The extracted memory-item title set changes for every complete pair. The mean token-set Jaccard distance is 0.734789. A deterministic post-ready strategy-slot audit maps the frozen titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification/completeness. All four complete pairs change their slot set. The slot-set Jaccard distances are 0.571, 0.400, 0.500, and 0.500 for tasks 21, 22, 23, and 25, respectively, with mean 0.493. This is a structural diagnostic over frozen titles, not an embedding-based semantic claim.

Provider failures for task 24 and task 47 are excluded from the paired analysis. Both are failure-label calls with `ArkResponseStateError`; neither produces assistant text or a raw memory SHA. Private response identifiers are retained and GET-only recovery finds no text. The four completed failure-label calls produce 777, 852, 1,113, and 1,296 output tokens. These receipts distinguish provider response-state failures from a later memory-parser rejection, while leaving arm-dependent completion bias unresolved.

B Prompt-Perturbation Control Details

The control set is disjoint from the F0 requests. Each fresh trajectory is written under both released reward modes. One semantic-preserving rewording is also evaluated within each mode. Table 4 reports the paired distances used in the frozen sign-flip test.

Table 4: Full fresh prompt-perturbation control. Between is the released success-versus-failure memory distance; Within averages the two same-mode rewording distances.

Task	Original outcome	Between	Within	$B_i - W_i$
26	failure	0.726	0.662	0.065
50	failure	0.668	0.599	0.069
51	failure	0.776	0.534	0.242
96	failure	0.682	0.549	0.134
48	success	0.716	0.656	0.059
49	success	0.573	0.594	-0.022
117	success	0.703	0.593	0.110
126	success	0.755	0.572	0.182
Mean		0.700	0.595	0.105

C Downstream Confirmatory Details

C.1 Fixed-state action distributions

The distributional action stress test retains all four intervention states from the action-reach experiment and uses eight rollouts per memory condition. The per-state success-memory versus failure-memory TV distances are 0.625 for task 22 and 0.375 for task 23. They are 0.125 for task 24 and 0.250 for task 26. Their mean is 0.34375. The frozen 100,000-draw permutation audit gives $p = 0.310527$.

C.2 Frozen terminal matrix

The terminal confirmatory experiment reuses every complete F0 pair and every future task frozen for the earlier fixed-evidence run. Source-memory tasks form $\{21, 22, 23, 25\}$. Future tasks form $\{164, 385, 387, 388\}$. The design therefore contains 16 source–future cells. Each cell receives eight fresh success-memory rollouts and eight fresh failure-memory rollouts. The policy model is Doubao Seed 2.0 Mini at temperature 0.2. All 256 requested calls complete with zero provider failures.

The primary statistic is the mean across cells of the absolute difference between success-memory and failure-memory terminal benchmark success rates. The support gate is frozen before calls at an effect floor of 0.15 together with a within-cell permutation $p < 0.05$. The permutation audit uses 100,000 draws with seed 20260822. The observed statistic is 0.15625 and the audit gives $p = 0.00074$.

Table 5: F2R1 confirmatory terminal success rates. S denotes success-label memory and F denotes failure-label memory.

Source	Future	S rate	F rate	$\lvert\Delta\rvert$
21	164	1.000	1.000	0.000
21	385	0.000	0.250	0.250
21	387	0.125	0.250	0.125
21	388	1.000	0.625	0.375
22	164	1.000	1.000	0.000
22	385	0.000	0.125	0.125
22	387	0.125	0.000	0.125
22	388	0.250	1.000	0.750
23	164	1.000	1.000	0.000
23	385	0.000	0.000	0.000
23	387	0.125	0.250	0.125
23	388	1.000	1.000	0.000
25	164	1.000	1.000	0.000
25	385	0.000	0.000	0.000
25	387	0.125	0.750	0.625
25	388	1.000	1.000	0.000
	Mean absolute difference			0.15625

C.3 HETEROGENEITY AND CORRUPTION-RATE DECOMPOSITION

For each frozen cell, the confirmatory run estimates paired conditional success probabilities. Eight of the 16 cells have zero observed success-rate contrast. Six have positive failure-memory minus success-memory effects and two have negative effects. The largest squared effects occur at source/future pairs 22/388 and 25/387; together they contribute 78.2051% of the total squared-effect mass. This sign and magnitude heterogeneity is why the revised paper treats the experiment as a forced-leverage measurement rather than a uniform directional effect or a native end-to-end transport estimate.

If reward-label corruption selects the failure-conditioned memory with probability q, the induced between-memory variance equals $q(1-q)\Delta_c^2$ in cell c. The exact mean squared conditional effect across all 16 cells is 0.076172. Averaging over cells gives the curve $0.076172q(1-q)$, with value 0.006855 at $q = 0.1$, value 0.014282 at $q = 0.25$, and value 0.019043 at $q = 0.5$. This remains a descriptive plug-in calculation conditional on forced memory exposure; it is not used as the main claim in the revised stage-resolved paper.

C.4 NATIVE TRANSPORT AUDIT ADDED AFTER THE FORCED-LEVERAGE STUDY

The later frozen native-pipeline audit changes the interpretation of the forced matrix. In Shopping, branch-specific source memory is retrieved in 125/172 held-out opportunities after bank expansion, but the mean absolute native terminal success difference is only 0.02083 with permutation $p = 0.4289$; 34/36 terminal cells are zero. The corresponding first-action total-variation statistic is 0.06944 with $p = 0.5801$, and no frozen state changes its modal first action between reward branches. A post-hoc working-memory branch-shift diagnostic is 0.00335 with $p = 0.2052$.

The cross-domain Reddit audit contains four complete success/failure writes, all of which diverge. Its native terminal mean absolute difference is 0.125 with $p = 0.2253$; 6/8 cells are zero and the two nonzero cells have opposite signs. These later measurements are why the main text separates forced capacity from native transport and treats attenuation after exposure as the strongest supported operational boundary rather than a causal mediation result.

For finite-cell sensitivity, we run a fixed-seed nonparametric bootstrap with 100,000 repetitions, resampling 16 source–future cells with replacement. The percentile 95% interval for mean absolute effect is $[0.054688, 0.273438]$. The corresponding interval for mean squared effect is $[0.010742, 0.164062]$, which maps to $[0.002686, 0.041016]$ for $V(0.5)$. This bootstrap is descriptive sensitivity to the observed cell composition. The preregistered within-cell permutation test on the 256 confirmatory rollouts supplies the primary significance result.